

TASK 2: AZURE CORE SOLUTIONS AND MANAGEMENT TOOLS

Chapter 07

Step 1 – Create a Resource Group

The screenshot displays the Microsoft Azure portal interface. The top navigation bar includes the 'Microsoft Azure' logo, an 'Upgrade' button, a search bar, and a 'Copilot' button. The user's profile 'ntibosiyabonga@gmail...' is visible in the top right corner.

The main content area shows the 'Create a resource group' wizard. The 'Basics' tab is selected, showing the following details:

- Subscription: Azure subscription 1
- Resource group name: RG-BCX-Siyabonga
- Region: (Africa) South Africa North

Below the wizard, the 'Tags' page for the 'RG-BCX-Siyabonga' resource group is displayed. The left sidebar contains a navigation menu with options: Overview, Activity log, Access control (IAM), Tags (selected), Resource visualizer, Events, and Settings. The 'Tags' page includes a search bar, 'Delete all' and 'Feedback' buttons, and a table of tags.

Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. Tag names are case insensitive, but tag values are case sensitive. [Learn more about tags](#)

Do not enter names or values that could make your resources less secure or that contain personal/sensitive information because tag data will be replicated globally.

Name	Value
Project	BCXClient
Owner	Siyabonga
Department	CloudOps
CostCenter	BCX001

At the bottom, a summary of the resource group is shown: RG-BCX-Siyabonga (Resource group). Below this, a list of tags is displayed: Project : BCXClient, Owner : Siyabonga, Department : CloudOps, and CostCenter : BCX001.

Tags are used to logically organize Azure resources for management, billing, and governance purposes. By applying Project, Owner, Department, and CostCenter tags, BCX can track resource ownership, assign costs, and enforce policies across all client resources

Step 2 – Deploy a Storage Account

The image shows two screenshots from the Microsoft Azure portal. The top screenshot is the 'Create a storage account' form. The bottom screenshot is the 'Overview' page for the deployment.

Create a storage account form:

- Subscription:** Azure subscription 1
- Resource group:** RG-BCX-Siyabonga
- Instance details:**
 - Storage account name:** stbcxsiyabonga
 - Region:** (Africa) South Africa North
 - Preferred storage type:** Azure Files
- Performance:** Standard (Recommended for general purpose file shares and cost sensitive applications, such as HDD file shares)
- File share billing:** Pay-as-you-go file shares: Billed based on usage
- Redundancy:** Locally-redundant storage (LRS)

Overview page:

Your deployment is complete

Deployment name : stbcxsiyabonga_1775652217318
Subscription : Azure subscription 1
Resource group : RG-BCX-Siyabonga

Start time : 4/8/2025, 2:45:43 PM
Correlation ID : fe32e891-96ae-4364-80aa-d9b92f008ffa

Resource	Type	Status	Operation details
stbcxsiyabonga/default	Microsoft.Storage/storageAcco...	OK	Operation details
stbcxsiyabonga	Storage account	OK	Operation details

Next steps: Go to resource

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Storage accounts provide reliable cloud storage for blobs, files, tables, and queues. BCX can store client data, logs, and backups securely.

Step 3 – Deploy a Virtual Machine (VM)

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Home > Compute infrastructure > Virtual machines

Create a virtual machine

Help me create a VM optimized for high availability Help me choose the right VM size for my workload Help me create a low cost VM

⚠ Changing Basic options may reset selections you have made. Review all options prior to creating the virtual machine.

Subscription * Azure subscription 1

Resource group * RG-BCX-Siyabonga
[Create new](#)

Instance details

Virtual machine name * VM-BCX-Siyabonga

Region * (Africa) South Africa North
[Deploy to an Azure Extended Zone](#)

Availability options Availability zone

Zone options

☒ Self-selected zone
Choose up to 3 availability zones, one VM per zone

☐ Azure-selected zone (Preview)
Let Azure assign the best zone for your needs

Availability zone * Zone 1

⚡ You can now select multiple zones. Selecting multiple zones will create one VM per zone. [Learn more](#)

Security type Trusted launch virtual machines

< Previous Next : Disks > Review + create Give feedback

https://portal.azure.com/?feature.token caching=true&feature.internalgraphapiversion=true#view/HubsExtension/DeploymentData...

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Home

CreateVm-canonical.ubuntu-24_04-lts-server-20260408144855 | Overview

Deployment

Search Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : CreateVm-canonical.ubuntu-24_04-lts-server-20260408144855 Start time : 4/8/2026, 2:52:59 PM

Subscription : Azure subscription 1 Correlation ID : 8849fd53-4e01-4d5d-8f64-a2ad40c9fd56

Resource group : RG-BCX-Siyabonga

Deployment details

Resource	Type	Status	Operation details
VM-BCX-Siyabonga	Virtual machine	OK	Operation details
vm-bcx-siyabongaaz19_21	Microsoft.Network/networkinter	OK	Operation details
VM-BCX-Siyabonga-ip	Public IP address	OK	Operation details
VM-BCX-Siyabonga-vnet	Virtual network	OK	Operation details
VM-BCX-Siyabonga-nsg	Network security group	OK	Operation details

Next steps

Set up auto-shutdown Recommended

Monitor VM health, performance, and network dependencies Recommended

Run a script inside the virtual machine Recommended

[Go to resource](#) [Create another VM](#) [Scale out your VM](#)

Add or remove favorites by pressing CTRL+SHIFT+F Give feedback

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill. [Set up cost alerts >](#)

This VM demonstrates IaaS (Infrastructure as a Service), allowing BCX to manage the OS and applications while Azure manages the underlying physical hardware.

Step 4 – IoT Service Selection Exercise

Requirement	Recommended Azure Service	Justification
Large-scale custom device connectivity	IoT Hub	Supports millions of devices, flexible protocols
Easy dashboard-driven management	IoT Central	Pre-built dashboards, no coding required
Enhanced device security	Azure Sphere	Hardware + OS security for devices
Edge / offline processing	IoT Edge	Run logic and AI locally at edge devices

Step 6 – Serverless Computing Selection

Requirement	Recommended Azure Service	Justification
Event-driven code execution	Azure Functions	Run small tasks triggered by events
Workflow orchestration	Azure Logic Apps	Connect apps and automate workflows visually
Event routing	Azure Event Grid	Route events between apps and services reliably

Step 7 – Monitoring Tool Selection

- **Azure Advisor:** Provides recommendations for cost, security, performance, and high availability
- **Azure Monitor:** Collects metrics and logs for VMs, apps, and resources
- **Azure Service Health:** Notifies about service outages or planned maintenance

Search resources, services, and docs (G+I)

Copilot

ntibosiyabonga@gmail....
DEFAULT DIRECTORY (NTIBOSIYA...

VM-BCX-Siyabonga
Virtual machine

Help me copy this VM in any region

Manage this VM with Azure CLI

Search

Help me copy this VM in any region

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Monitor

Resource visualizer

Connect

Connect

Bastion

Networking

Network settings

Load balancing

Application security groups

Network manager

Settings

Availability + scale

Security

Connect

Start

Restart

Stop

Hibernate

Capture

Delete

Refresh

Scale

Essentials

JSON View

Resource group (move)
RG-BCX-Siyabonga

Status
Running

Location
South Africa North (Zone 1)

Subscription (move)
Azure subscription 1

Subscription ID
fbfd5a28-8253-460d-850b-28eb0df81850

Availability zone
1

Operating system
Linux (ubuntu 24.04)

Size
Standard D2s v3 (2 vcpus, 8 GiB memory)

Primary NIC public IP
20.164.18.18
1 associated public IPs

Virtual network/subnet
VM-BCX-Siyabonga-vnet/default

DNS name
Not configured

Health state
-

Time created
4/8/2026, 12:53 PM UTC

Tags (edit)
Add tags

Properties

Monitoring

Capabilities (7)

Recommendations

Tutorials

Virtual machine

Computer name
VM-BCX-Siyabonga

Networking

Public IP address ⓘ
20.164.18.18 (Network vm-bcx-)

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